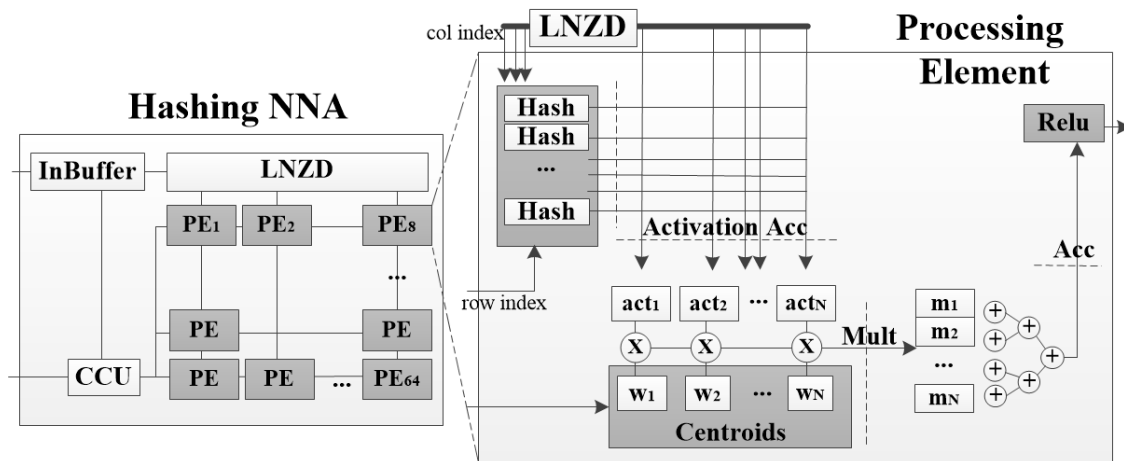


VLSI Implementation of Area-Efficient Parallelized Neural Network Accelerator Using Hashing Trick

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- **Abstract**

- Implemented custom Neural Network Accelerator(NNA)
 - Improved area-performance efficiency
 - Used hashing trick quantization



- **0. Background**
- **1. Design Exploration**
- **2. Experiment**
- **3. Conclusion**

- **Neural Network Accelerator(NNA)**
 - Vector processor : accelerating Neural Network
 - ASIC, FPGA : Mainly for performance (ex) TPU
 - Recent focus : for embedded devices
 - Constraint resources (area, memory, power dissipation)
 - This paper : **employ Weight Sharing Quantization - hashing trick**

Name	EIE (ISCA'16)	Cambricon-X (MICRO'16)	Cnvlutin (ISCA'16)	Eyeriss (ISCA'16)	SCNN (ISCA'17)	TPU (ISCA'17)	Sparsecore (sysml'18)
Target	FC	FC/Conv	Conv	Conv	Conv	FC/Conv	Conv
PE type	SIMD	SIMD	SIMD	Systolic	Systolic	Systolic	SIMD
Sparse	Interleaved CCS format	Step indexing	ZFNAF format	X	Interleaved CCS format	X	CCS format
Quantization	Weight Sharing	Low bit(16bit)	Low bit (16bit)	X	Low bit(16bits)	X	X

0. Background

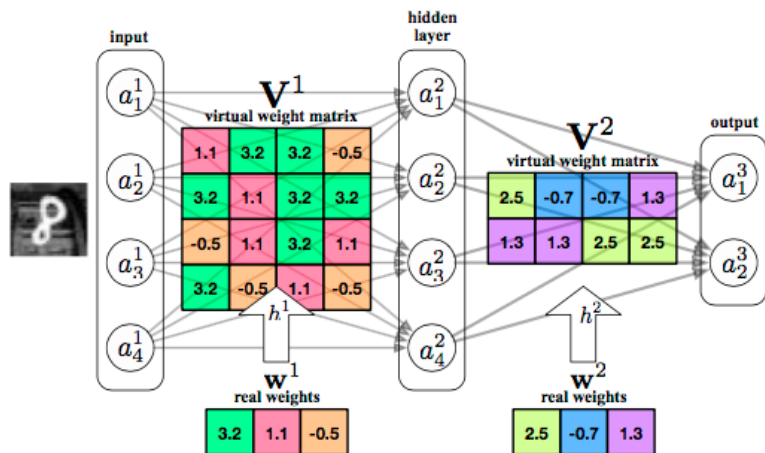
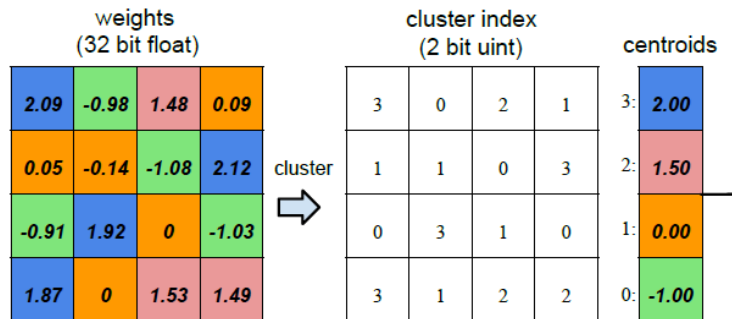
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- Quantization

- 1. Parameter sharing
 - 2. Low precision

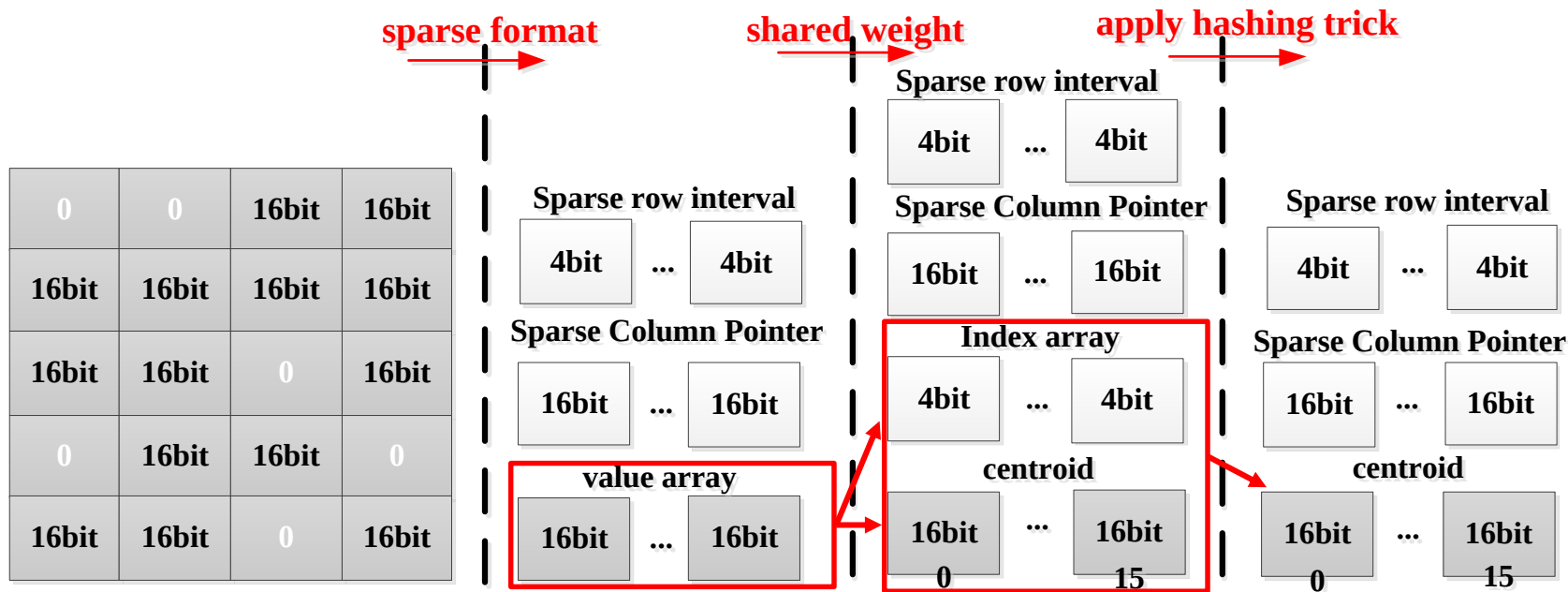
- Parameter sharing examples

- Deep Compression (ICLR'16)
 - Employed model in **EIE** (ISCA'16)
 - K-means clustering
 - Index + centroids
 - Hashing Trick (ICML'16)
 - Random weight sharing



- **Data Transition**

- Hashing trick : minimum information of weights



- **Problem of previous accelerators**
 - “The **on-chip buffers** consume significant chip area(up to 70%), greatly increasing the cost and decreasing the computational density”
 - *TETRIS(OSR'17)*
 - “The area(93%) and power(59%) is dominated by **SRAM**”
 - *EIE(ISCA'16)*
 - That is, area-performance efficiency was not considered
 - The problem of SRAM size
- **Objective**
 - Minimize SRAM for area-efficient NNA
 - How to reduce needed weights more?
 - Weight sharing
 - This paper : Hashing trick

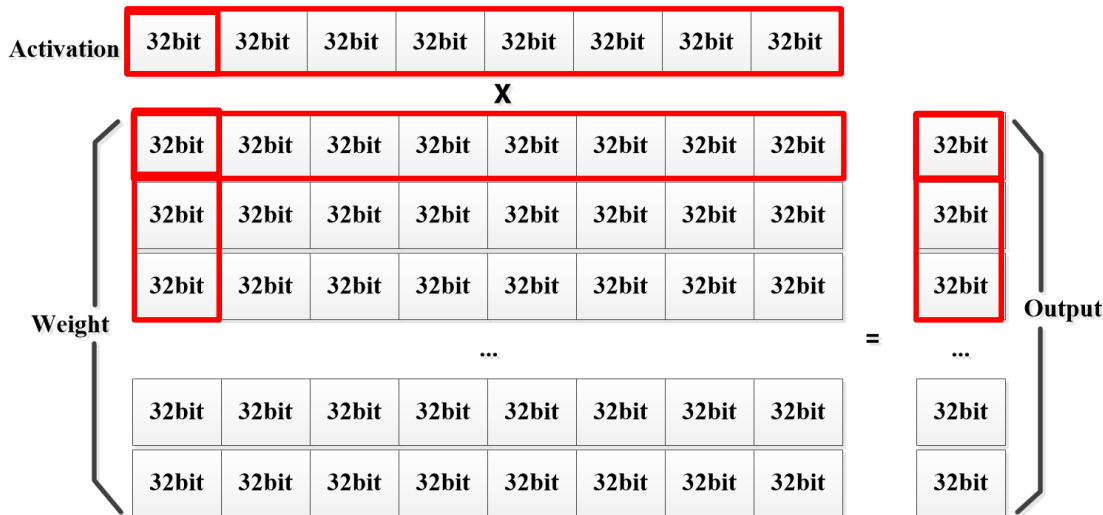
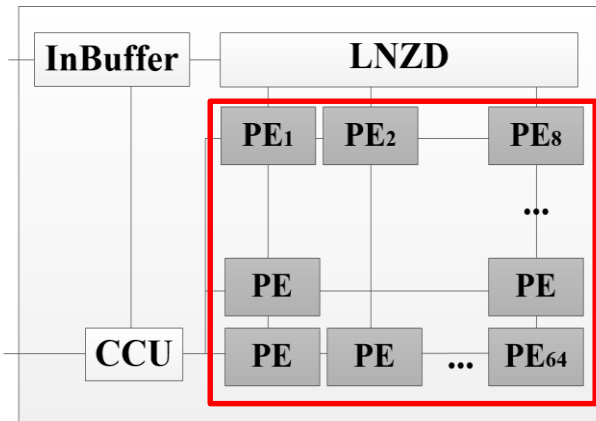
1. Design Exploration

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- **Design Exploration for PE (1/2)**

- 1. PARALLEL : parallelism in PE
- 2. SKIP : employ sparsity (weight or activation)
 - Sometimes, restricted to employ both of sparsity

Hashing NNA

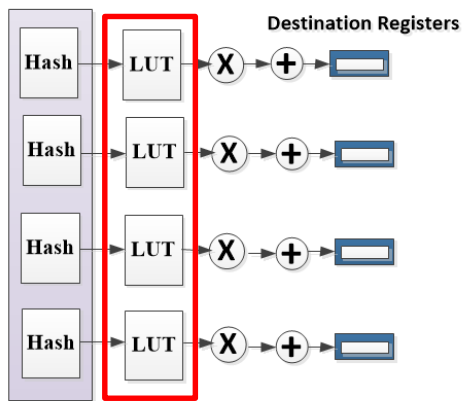


1. Design Exploration

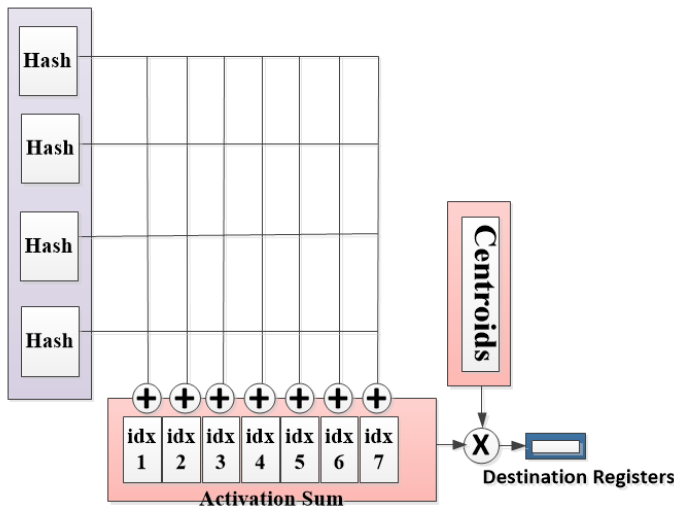
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- **Design Exploration for PE**
 - 1. PARALLEL : Weight sharing problem
 - Many Look Up Tables (SRAM size $\uparrow$)
 - Suggested **post-multiplication method**

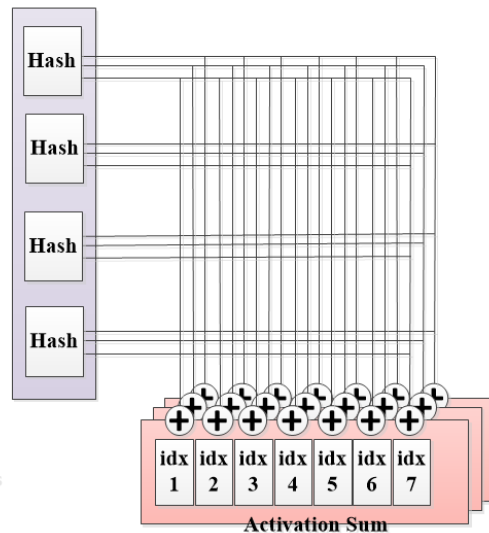
<Parallel without post-mult>



<one output : Possible>



<three outputs : Impossible>



1. Design Exploration

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- **Design Exploration for PE (2/2)**
 - 1. PARALLEL
 - LUT problem solved in D4
 - 2. SKIP

	PARALLEL(A+W)		SKIP(A+W)		Post-multiply	Disadvantage
	A	W	A	W		
D1	Serial	Serial	skip	skip(CSC)	Impossible	Reduce 36% of SRAM size, but area↑
D2	Serial	Parallel(↑)	skip	skip(CSC)	Impossible	LUT problem(SRAM size↑)
D3			skip	X	Impossible	LUT problem(SRAM size↑)
D4	Parallel(↔)	Parallel(↔)	skip	X	Possible	LUT problem → solved
D5			X	skip(CSR)	Possible	LUT problem → solved, but still 36%

2. Experiment

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- **Result Design (1/2)**

- Implemented with Verilog

- Synthesized with ***Design Compiler*** under 32nm CMOS tech

- Area was estimated with ***Synopsys IC Compiler***

- Operated at 200MHz

- Result comparison

- N=64 takes only **1.86mm²**
more than N=32 : scalable

- More than expected area
23.8mm² (N=64)
because of interconnection

N = the number of PEs

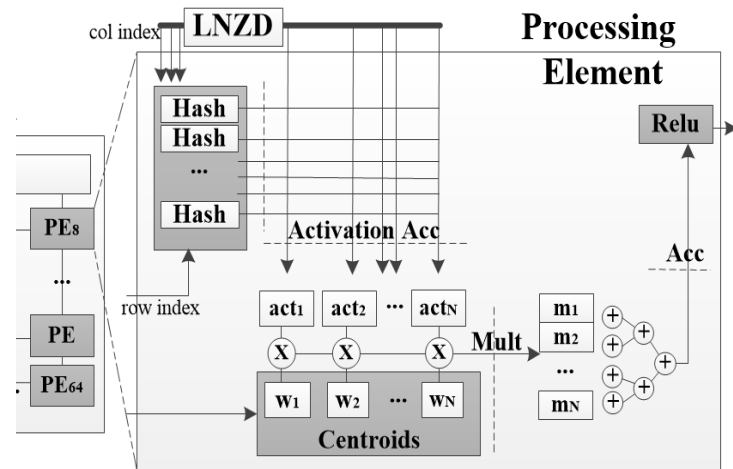
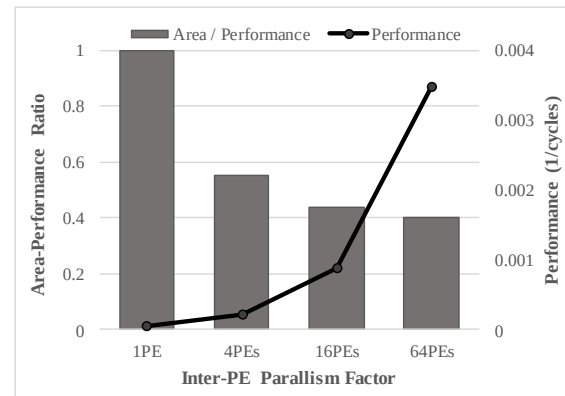
			N = 32	N = 64
Critical path delay			4.85 ns	4.87 ns
SRAM size			16 KB	16 KB
SRAM area			0.76 mm ²	0.76 mm ²
1PE	Logic gate count		388,955	414,109
	Area	Logic	0.46 mm ²	0.49 mm ²
		Total	1.12 mm ²	1.25 mm ²
64PE	Logic gate count		25,671,236	27,595,669
	Area	Logic	29.5 mm ²	31.3 mm ²
		Total	30.2 mm ²	32.06 mm ²

2. Experiment

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• Result Design (2/2)

- Compared with EIE(MNIST)
 - PE : 64(EIE)
- Performance
 - For matrix 784 x 1000
 - Assumed sparsity : W(0.9), ACT(0.7)
 - PRALLEL : 16
 - **37.5%** cycles ↓ = **60%** performance ↑
- Area/SRAM size
 - SRAM size : 656KB → 16KB (**97.56%** ↓)
 - Area : **13.9%** area ↓
- Performance-area efficiency
 - **60%** ↑ / **13.9%** ↓ = **85.83%** ↑



- **Future work**
 - Original Hashed-Net : XXhash logic **(Done)**
 - Not optimized for hardware
 - Too many **multiplications**
 - → **XOR** operation
 - Showed almost **no accuracy degradation**
 - Accuracy Compensation
 - Problem of hashing trick : Random mapping
 - Accuracy compensation needed
 - Different training skill for BIG dataset
- **Contribution**
 - Improved area-performance efficiency
 - More scalable & Suitable for embedded device

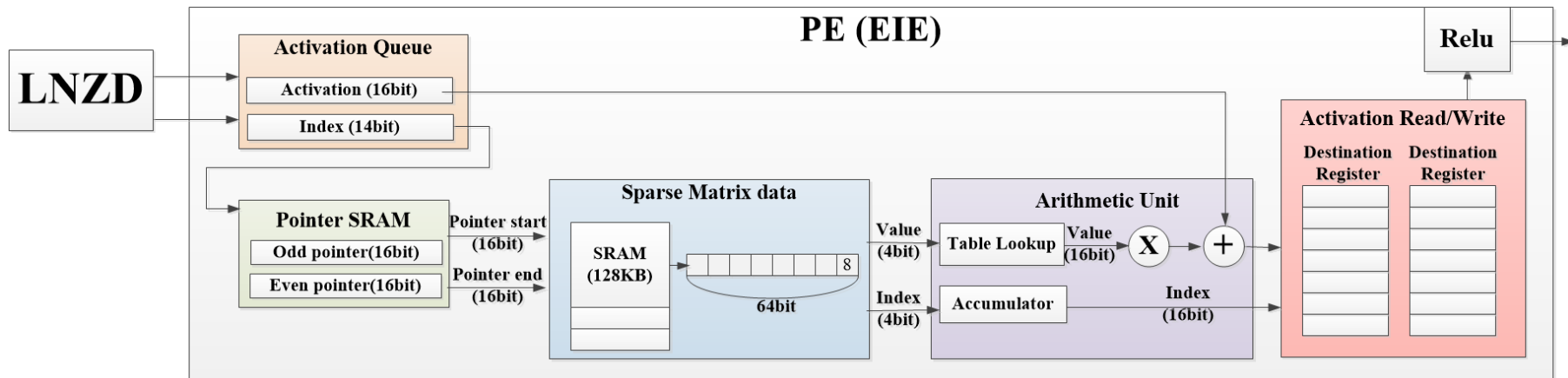


- **EIE(ISCA'16)**

- Put all parameters in on-chip SRAM

- Using **Deep Compression(ICLR'16)**

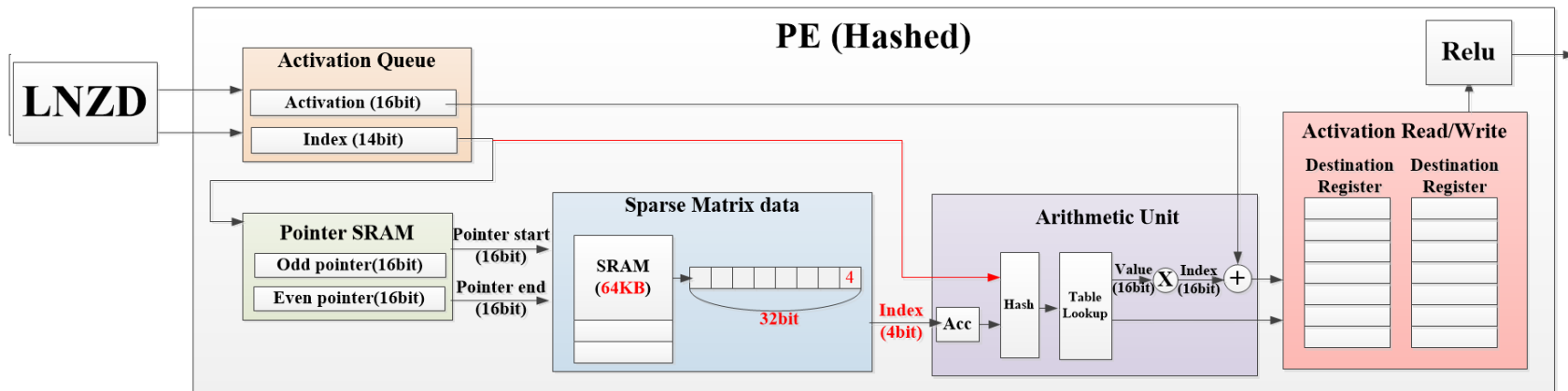
- 1. Sparsity : Pruning
- 2. Shared weight : **Quantization(parameter sharing)**
- 3. Huffman coding



Appendix0. EIE (2/2)

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- **Simplest design**
 - **EIE SRAM**
 - 10752KB : 2048KB(Ptr) + 8192KB(SPmat) + 128KB(Act) + LUT(512KB)
 - **Hashing trick SRAM**
 - 6784KB(36.9%↓) : 2048KB(Ptr) + 4096KB + 128KB(Act) + LUT(512KB)
 - Area not much influenced : **Hash unit**



Appendix1. Accelerators (1/2)

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- Accelerators

Name	EIE (ISCA'16)	Cambricon-X (MICRO'16)	Cnvlutin (ISCA'16)	Eyeriss (ISCA'16)	SCNN (ISCA'17)	TPU (ISCA'17)	Sparsecore (sysml'18)
Target	FC	FC/Conv	Conv	Conv	Conv	FC/Conv	Conv
Dataflow	Inner Product (A+W)	Inner Product (W)	Vector Scalar + Reduction (A)	Row stationary	Cartesian Product + Input stationary (A+W)		Vector Scalar (A+W)
PE type	SIMD	SIMD	SIMD	Systolic	Systolic	Systolic	SIMD
Sparse	Interleaved CCS format	Step indexing	ZFNAF format	X	Interleaved CCS format	X	CCS format
Quantization	Weight Sharing	Low bit(16bit)	Low bit (16bit)	X	Low bit(16bits)	X	X

Appendix1. Accelerators (2/2)

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- **EIE(ISCA'16)** : sparsity + quantization* + Huffman-coding
- **Cambricon-X(MICRO'16)** : sparsity of weight
- **Cnvlutin(CAN'16)** : sparsity of activation
- **Eyeriss(ISCA'16)** : Row-stationary (Novel dataflow) for CONV
- **SCNN(CAN'17)** : sparsity of weight & activation
- **TPU(ISCA'17)** : Google, 30x and 15x faster than CPU & GPU
- **Sparsecore(Sysml'18)** : k/m sparsity for being fully utilized

- **Activation parallel skip problem**

- LNZD(Leading Non-Zero Detection)

- Bandwidth : for 16bit

- 16 activations : 512bits

- 32 activations : 1024bits

- Snapdragon 855 (31.79GB/s)

- Bandwidth : 31.79 GB/s

- » 512bits : 16 ns

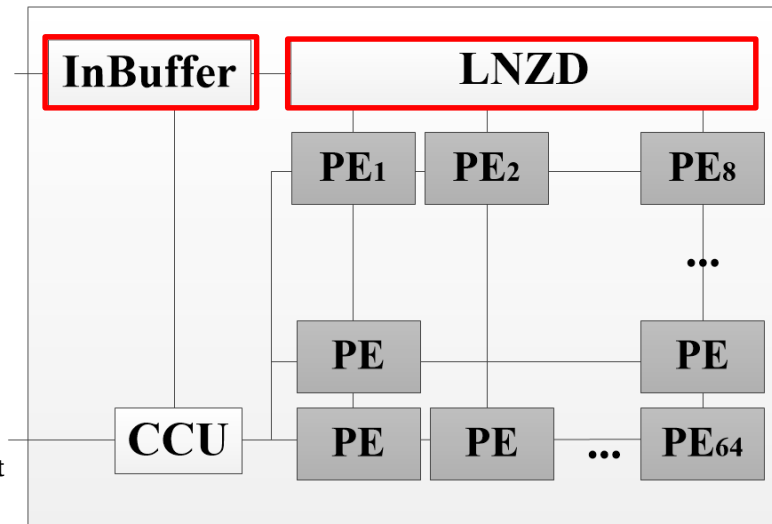
- » 1024bits : 32 ns

- 200Hz = 0.005s/cycle

- » 512bits :

CCU : Central Control Unit
PE : Processing Element

Hashing NNA



Appendix3. Alternative hash (1/1)

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- **Non-cryptographic hash function**

- No multiplication, No table

- **Jenkins** hash function

- XOR, addition

- OAT hash function

- One At a Time

- Just under 100%

- Buzhash function

- Avoiding multiplication using barrel shifts

- Trained with new seed value

- Error : test (0.0172), valid(0.0168)

